3 The trends reshaping cybersecurity

As organizations confront AI threats, geopolitical volatility and supply chain vulnerabilities, the need for resilience has never been clearer.



3.1 AI is reshaping risk, accelerating both offence and defence

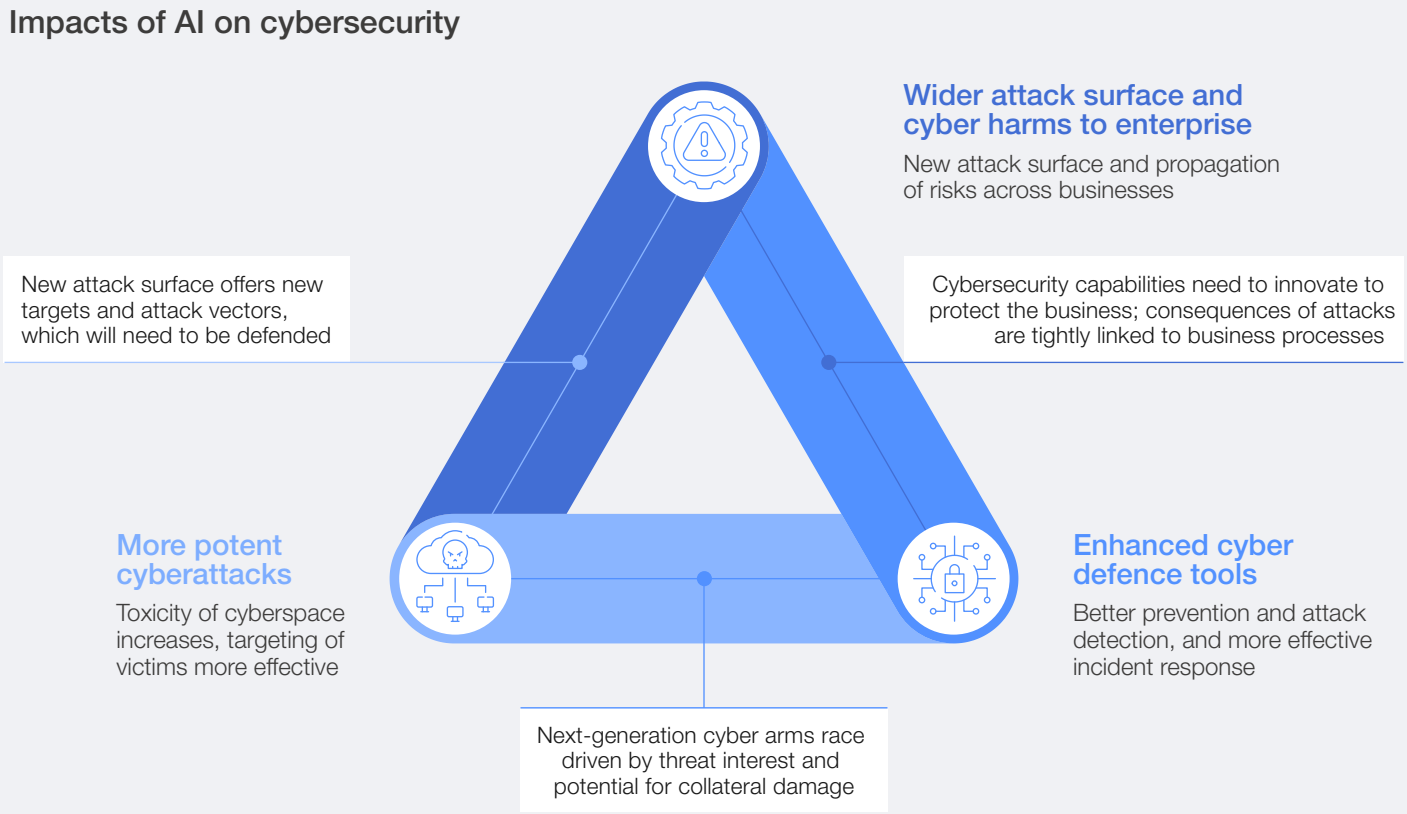
Developments in AI are reshaping multiple domains, including cybersecurity. Implemented well, these technologies can assist and support human operators in detecting, defending and responding to cyberthreats. However, they can also pose serious risks such as data leaks, cyberattacks and online harms if they malfunction, or are misused. Governments must take a forward-looking, practical and collaborative approach to developing and using emerging technologies safely, as their capabilities and risks continue to evolve. The risks transcend borders, and the challenge is to maximize AI's benefits, including to strengthen our cyber resilience, while minimizing its risks.

Josephine Teo, Minister for Digital Development and Information and Minister-in-Charge of Cybersecurity and Smart Nation Group, Singapore

AI is anticipated to be the most significant driver of change in cybersecurity in the year ahead, according to 94% of survey respondents. AI is reshaping the cybersecurity landscape across three interconnected dimensions. First, the widespread integration of AI systems introduces an expanded attack surface, creating novel vulnerabilities that traditional controls were not designed to address. Second, defenders are harnessing AI to strengthen their cyber capabilities – augmenting detection, accelerating incident response and automating high-volume analytical tasks. Third, threat actors are leveraging AI to enhance the scale, speed, sophistication and precision of their attacks, driving a new generation of automated exploitation and targeted social engineering (see Section 3.3).

Together, these dynamics illustrate the dual-use nature of AI, both as a force multiplier for defence and as a catalyst for attackers. As this technological competition intensifies, organizations are shifting from reactive to proactive security, while reassessing governance, validation and oversight at every stage of AI adoption.

FIGURE 7 **Impacts of AI on cybersecurity**



Source: Artificial Intelligence and Cybersecurity: Balancing Risks and Rewards. (2025). World Economic Forum[2]

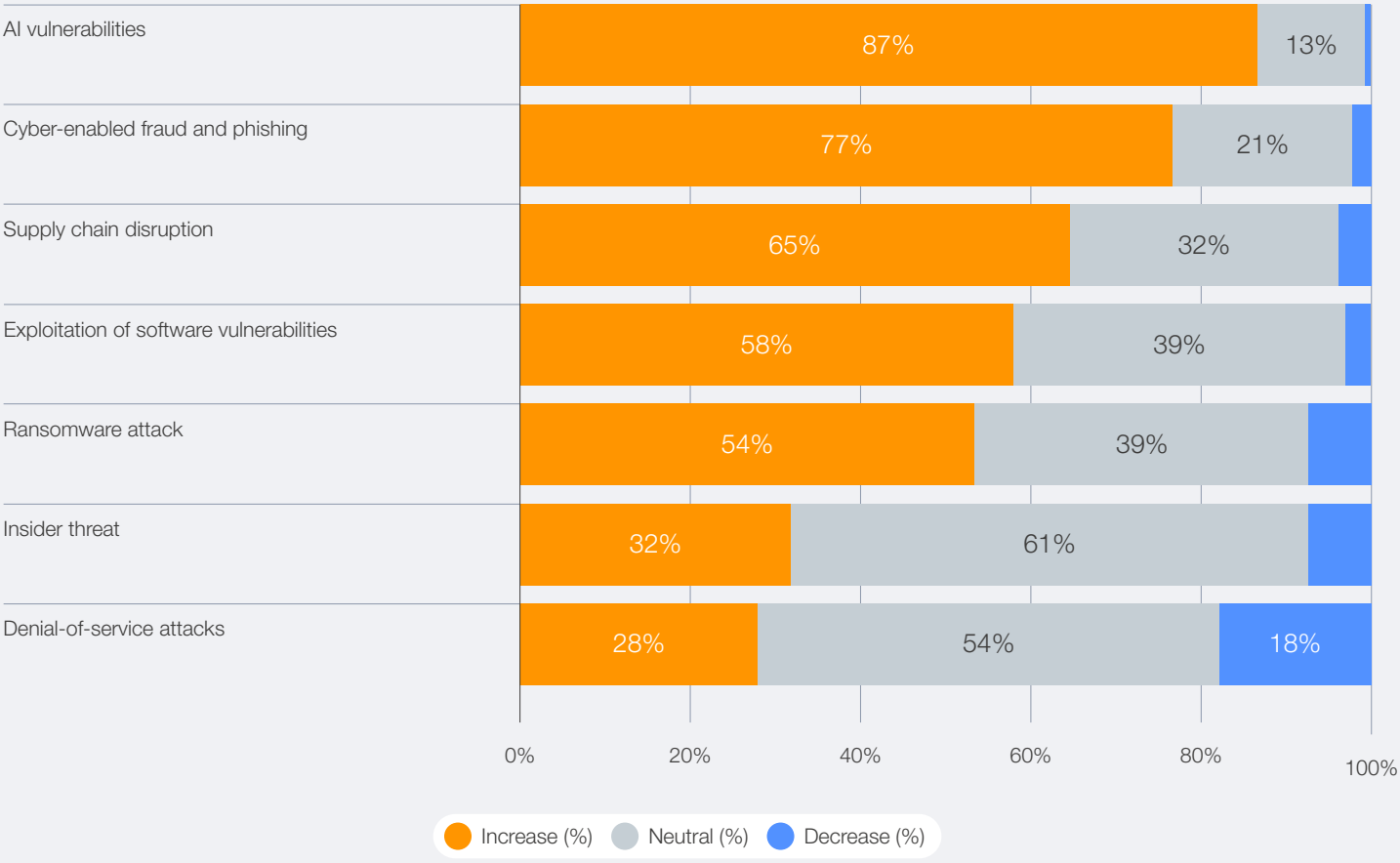
AI's benefits are contingent on disciplined execution. Poorly implemented solutions can introduce new risks – misconfiguration, biased decision-making, over-reliance on automation and susceptibility to adversarial manipulation – unless organizations embed robust guardrails, security-by-design practices and continuous monitoring. The implication is clear: AI can improve cybersecurity, but only when deployed within sound governance frameworks that keep human judgement at the centre. At the same time, too many controls can create friction, so it is essential to strike a careful balance.

Security of AI: from awareness to action

According to the *Global Cybersecurity Outlook 2026* survey, 87% of respondents identified AI-related vulnerabilities as the fastest-growing cyber risk over the course of 2025.

FIGURE 8 **Perception of increase or decrease in cyber risks over the past year**

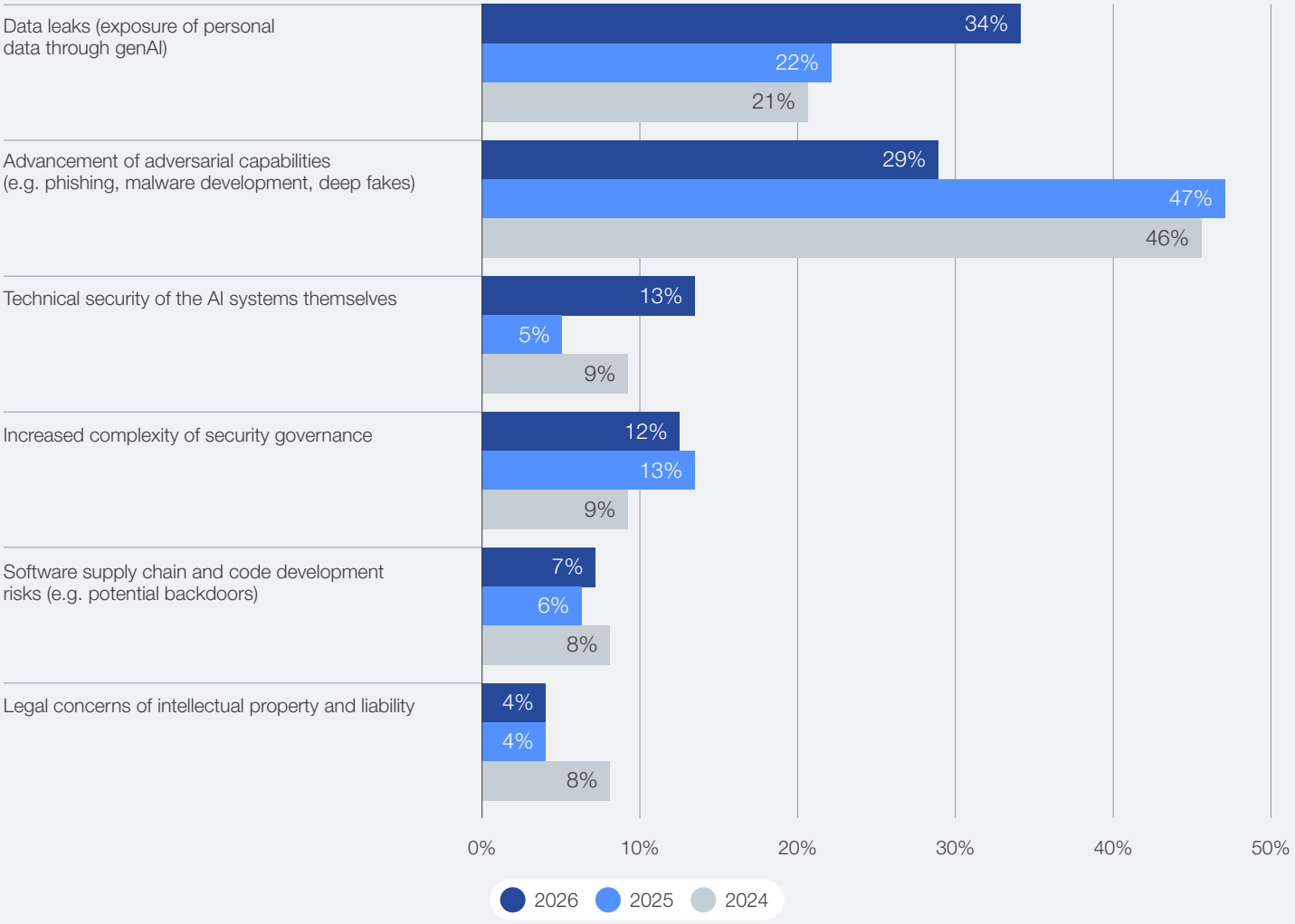
In the past year, do you think the following cyber risks have increased, decreased or stayed the same?



Data leaks associated with genAI (34%) and the advancement of adversarial capabilities (29%) stand out as leading concerns for 2026. This marks a striking reversal from previous years – in 2025, advancement of adversarial capabilities topped the list at 47% compared to only 22% for data leaks associated with genAI. The shift underscores a turning point in the AI risk landscape for the upcoming year: while the "AI arms race" between attackers and defenders continues to intensify, attention is pivoting from purely offensive innovation with AI towards the unintended exposure and misuse of sensitive data through generative and agentic systems.

FIGURE 9 | **Top concerns related to genAI**

Which cybersecurity issues related to generative AI concern you the most?



Growing awareness of AI-related cybersecurity risks is reflected in the increasing focus on the secure use of AI. The *Global Cybersecurity Outlook 2025* highlighted a significant gap between the widespread recognition of AI-driven risks and the rapid adoption of AI technologies without adequate safeguards. By 2026, however, this picture is changing: the share of organizations assessing the security of their AI tools has nearly doubled – from 37% in 2025 to 64% in 2026 – indicating that more organizations are introducing structured processes and governance models to manage AI securely and responsibly.

FIGURE 10 | **Percentage of organizations with processes in place to assess AI security**

Does your organization have a process in place to assess the security of AI tools before deploying them?

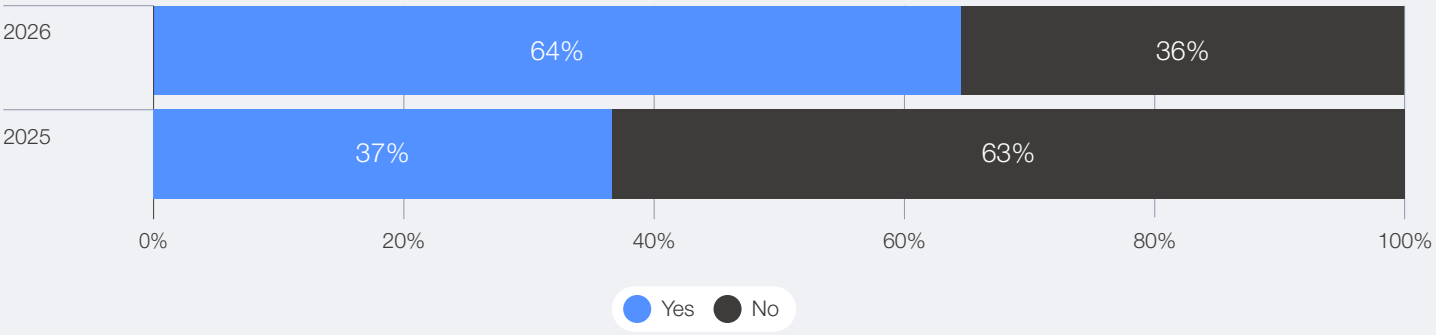
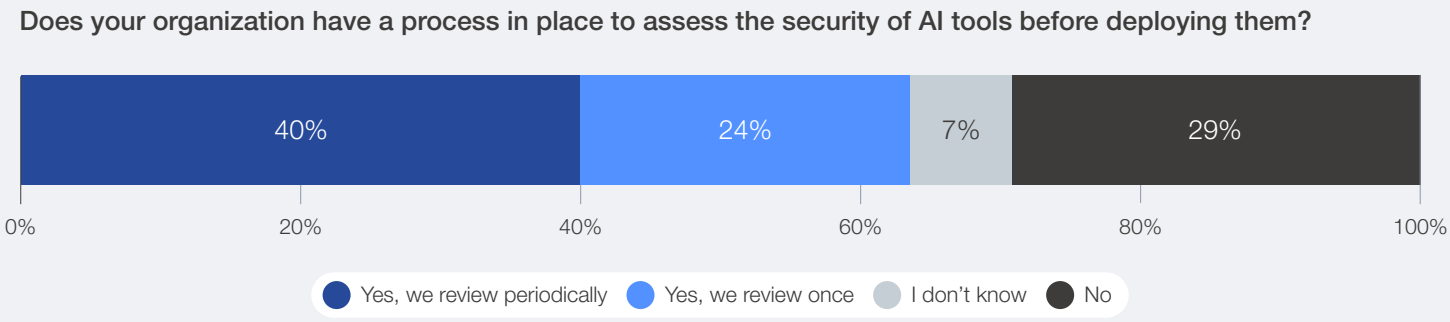




FIGURE 11 **Frequency of AI security assessments in organizations**



In the 2026 survey, 40% of organizations reported conducting periodic reviews of their AI tools before deploying them, rather than only doing a one-time assessment (24%) – a clear sign of progress towards continuous assurance. However, roughly one-third still lack any process to validate AI security before deployment, leaving systemic exposures even as the race to adopt AI in cyber defences accelerates.

The market's drive to adopt new AI features often outpaces security readiness, creating exploitable vulnerabilities.[3] In response to these emerging risks, a number of fundamental measures should be prioritized to secure AI at the infrastructure level. This includes protecting the data used in the training and customization of AI models from breaches and unauthorized access. AI systems should be developed with security as a core principle, incorporating regular updates and patches to address potential vulnerabilities. It is also critical for organizations to deploy robust authentication and encryption protocols to ensure the protection of customer interactions and data.[4]

BOX 1 **The widespread adoption of AI agents**

As AI agents become more widely adopted, they are poised to transform how digital systems are designed and developed. AI agents can enhance efficiency, responsiveness and scalability by automating complex or repetitive activities with speed and consistency, but their integration can challenge traditional security frameworks, redefining roles and processes, while raising fundamental questions about decision-making and the prioritization of alerts.

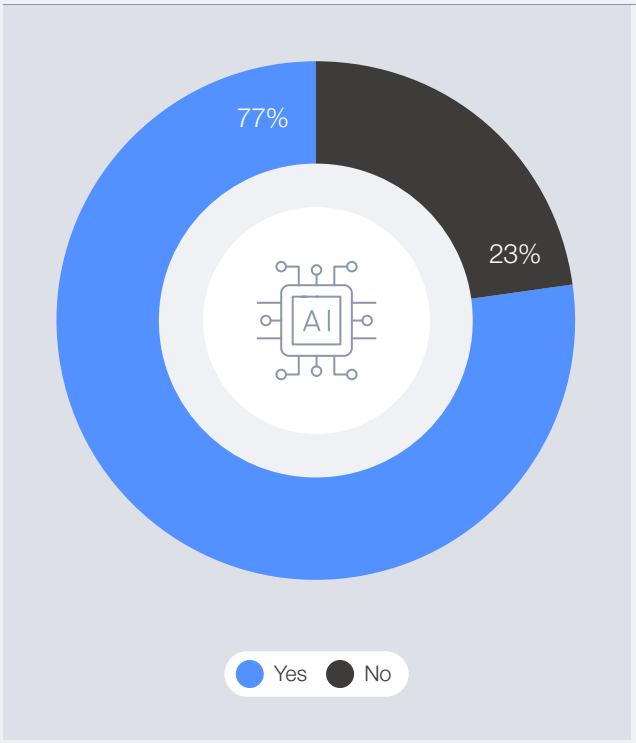
The multiplication of identities and connections makes managing their credentials, permissions and interactions just as critical – and likely even more complex – as managing those of human users. As outlined in the World Economic Forum's report *AI Agents in Action: Foundations for Evaluation and Governance*, without strong governance, agents can accumulate excessive privileges, be manipulated through design flaws or prompt injections, or inadvertently propagate errors and vulnerabilities at scale. Their speed and persistence amplify these risks, underscoring the need for continuous verification, audit trails and robust accountability structures grounded in zero-trust principles that treat every interaction as untrusted by default.[5]

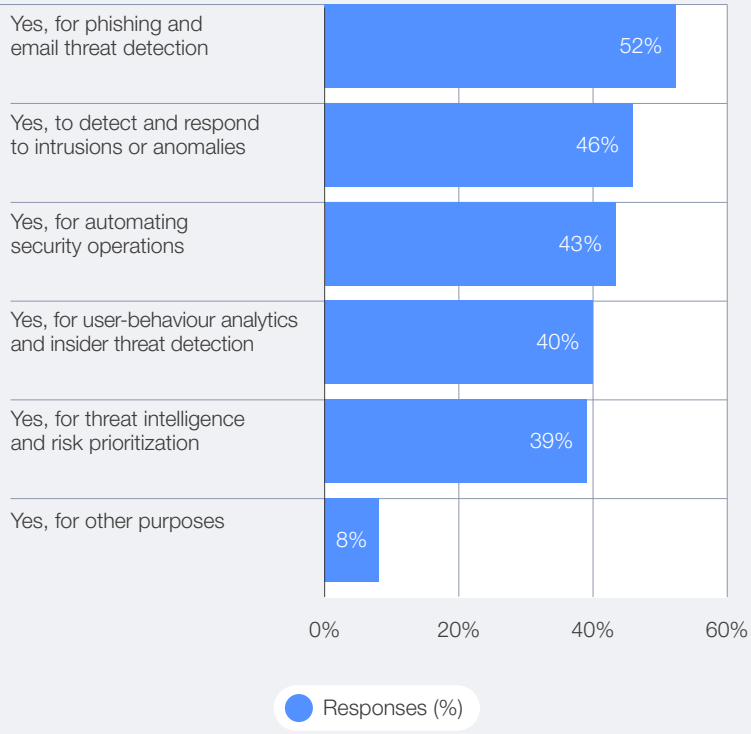
AI for cybersecurity

AI is fundamentally transforming security operations – accelerating detection, triage and response while automating labour-intensive tasks such as log analysis and compliance reporting. AI's ability to process vast datasets and identify patterns at speed positions it as a competitive advantage for organizations seeking to stay ahead of increasingly sophisticated cyberthreats. The survey data reveals that 77% of organizations have adopted AI for cybersecurity, primarily to enhance phishing detection (52%), intrusion and anomaly response (46%) and user-behaviour analytics (40%).

FIGURE 12 **How organizations are implementing AI for cybersecurity**

Has your organization implemented any AI-enabled tools to fulfil its cybersecurity objectives? (select up to three)





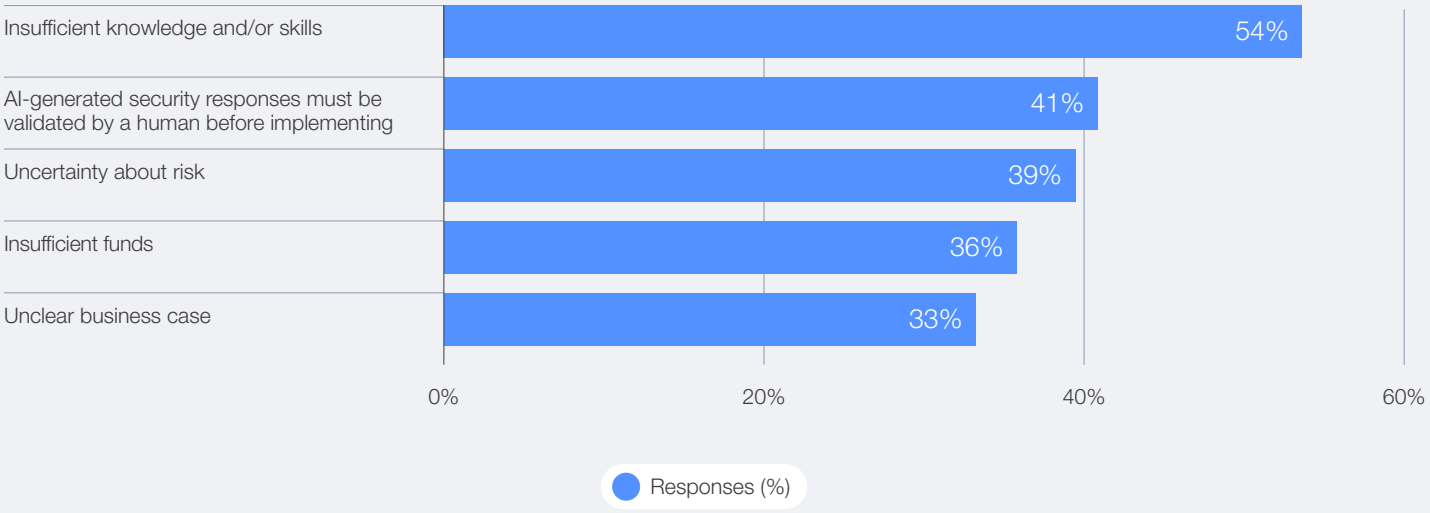
Addressing the practical challenges of AI adoption in cybersecurity, organizations consistently identify insufficient knowledge and/or skills (54%) to deploy AI for cybersecurity, the need for human oversight (41%) and uncertainty about risk (39%) as the main hurdles. These findings indicate that trust is still a barrier to widespread AI adoption.

> **Criminals are always willing to use all possible ways to get access to value, much of which is contained in the cyber infrastructure. Consequently, to stay ahead, those of us who defend must use every tool at our disposal – which now includes agentic AI.**
>
> Arvind Krishna, Chief Executive Officer, IBM



FIGURE 13 | **Barriers in AI implementation for cybersecurity**

What implementation hurdles does your organization face in embracing AI for cybersecurity? (select all that apply)



As organizations navigate the integration of AI into their security operations, the balance between automation and human judgement becomes increasingly critical. While AI excels at automating repetitive, high-volume tasks, its current limitations in contextual judgement and strategic decision-making remain clear. Over-reliance on ungoverned automation risks creating blind spots that adversaries may exploit.

This evolving dynamic is reshaping the role of cybersecurity professionals, highlighting the importance of adapting skill sets to meet new demands. According to the World Economic Forum's *The Future of Jobs Report 2025*, "networks and cybersecurity" are among the top three fastest-growing skills projected for 2030 – alongside AI and big data and technological literacy – reinforcing the urgency for targeted upskilling and continuous learning.[6]

Rather than replacing human expertise, AI is enabling specialists to shift their focus towards strategic oversight, governance and policy while delegating routine operational tasks to automation. This transition demands new skill sets, blending technical proficiency with strategic and ethical considerations, and underscores the growing importance of AI literacy across security teams.

The priorities for organizations are clear: invest in AI literacy and secure-use skills, and embed governance and validation, without creating new single points of failure. A collaborative model, anchored in security-by-design principles, emerges as the recommended path forward – enabling organizations to harness AI's advantages while mitigating vulnerabilities and ensuring innovation strengthens, rather than compromises, cybersecurity.

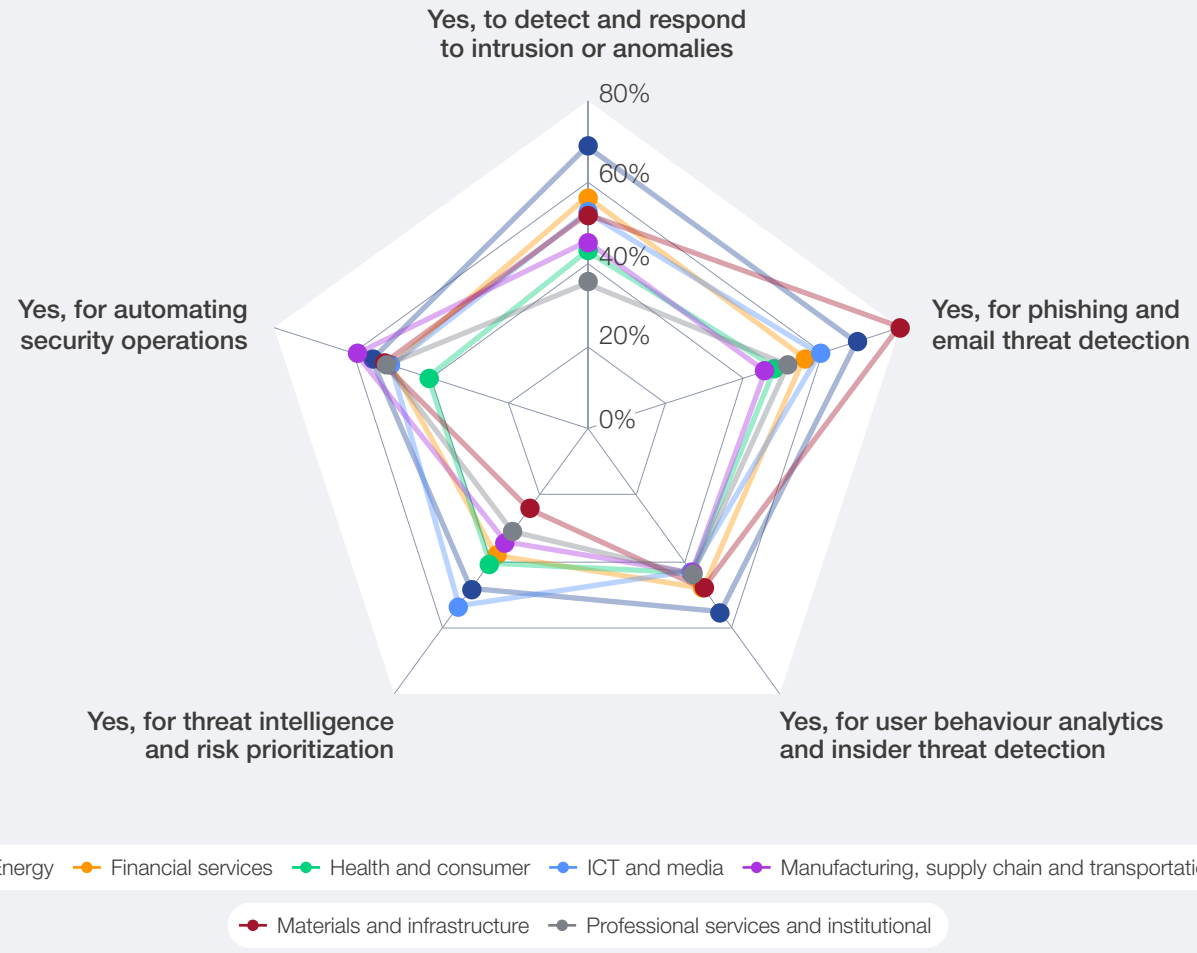
How industries are adopting AI for cybersecurity

The adoption of AI tools to augment cybersecurity capabilities varies across industries, reflecting sector-specific risk profiles and operational needs. The energy sector emphasizes intrusion and anomaly detection (according to 69% of respondents who have implemented AI for cybersecurity); the materials and infrastructure sector prioritizes phishing protection (80%); and the manufacturing, supply chain and transportation sector reports greater use of automated security operations (59%). These differences not only reflect sectoral risk profiles and operating constraints but also collectively point to a maturing portfolio of AI-enabled cyber defence capabilities that spans detection, intelligence, analytics and orchestrated cyber defence.

The differences in AI adoption for cybersecurity will be analysed in Section 3.6.

FIGURE 14 | **Use cases of AI for cyber defence across industries**

Has your organization implemented any AI-enabled tools to fulfil its cybersecurity objectives?



3.2 Geopolitics is a defining feature of cybersecurity

In an increasingly fragmented global environment – marked by conflicts, geoeconomic tensions, trade wars, sanctions and growing technological competition – geopolitics has become a defining force shaping cybersecurity.

The *Global Cybersecurity Outlook 2026* survey data reveals that, although the percentage of organizations changing their cybersecurity strategy due to geopolitics has declined from 93% in 2023 to 66% in 2026, geopolitics remains the top factor influencing overall cyber risk mitigation strategies. This suggests that the initial wave of adaptations following the geopolitical turmoil that dominated the headlines in 2022 and 2023 has passed, and that geopolitical risk is now a major factor shaping cyber defence.

